Build guide 2: molds and casting

DIY CNC epoxy-granite mill · October 2026

2. Molds and cores

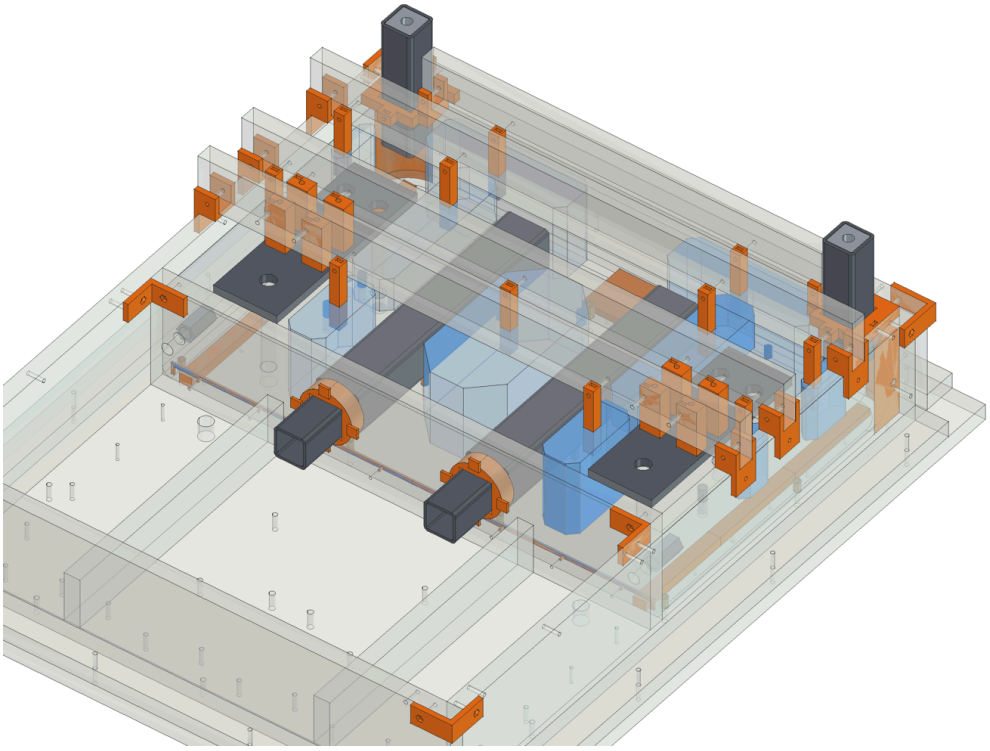
The molds are a kit. Melamine boards are cut to a list and drilled from hole maps, and printed brackets hold them together with M4 bolts. Every printed piece bolts or pins to its board from outside, into a nut built into the piece. Nothing is glued, nothing goes through a cast face, and the molds come apart and go back together the same way. The **mold guide** has the cut list, the hardware, a drawing and a hole table for every board, and the steps (`docs/3-molds/`, one guide per mold); the print guide lists the printed parts.

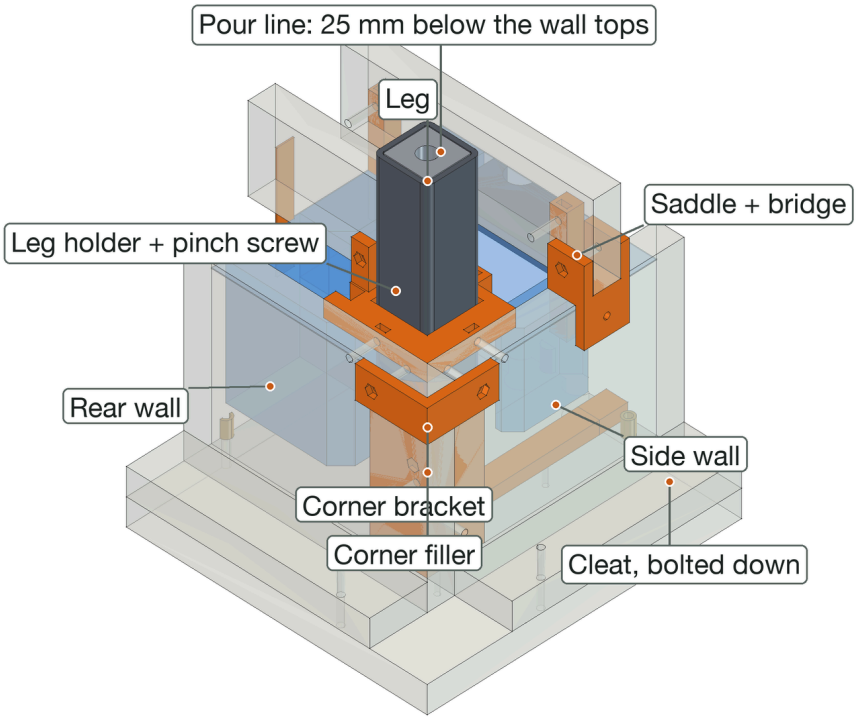
- **Walls stand 25 mm above the pour line.** The brackets, saddles and leg holders bolt through that band. Fill to the line, not to the top.
- **Everything that has to be flat or located goes against the floor**, so each casting is poured with its most important face down.
- **Edges:** the castings' outside edges are rounded 10 mm. Glue 3/8 in quarter-round molding (PVC or primed wood) into the inside corners of the boxes that form outside edges, and run small silicone fillets everywhere else.
- **Release:** wax every surface the mix touches 3-5 times (Partall Paste #2), buff, then spray or wipe PVA. That includes the printed pieces that come out; not the ones that stay in.

Casting	Face on the mold floor	Size (mm)	Mass
Base front	top face (coolant floor)	600 x 345 x 110	30.5 kg
Base rear	top face	600 x 380 x 110	41.9 kg
Upright (x2)	foot (the face that sits on the base)	80 wide, foot 400 long, 246 tall, sloped front	15.7 kg each
Beam	front face (rails)	530 x 140 x 140	26.3 kg

Casting order:

1. **Base rear half**, against a partition at the split, on a floor board big enough for both halves.
2. **Base front half** on the same floor, with the partition out, cast against the rear half's waxed split face. The two top faces come out flush and the split faces match. The front tubes and their sleeves slide straight onto the rear half's spigots, and the draw rods screw into its couplers, so that joint fits by construction. The split's gasket groove is in the rear half; tape over it first.
3. **Uprights**, both from one batch, in two molds.
4. **Beam.**

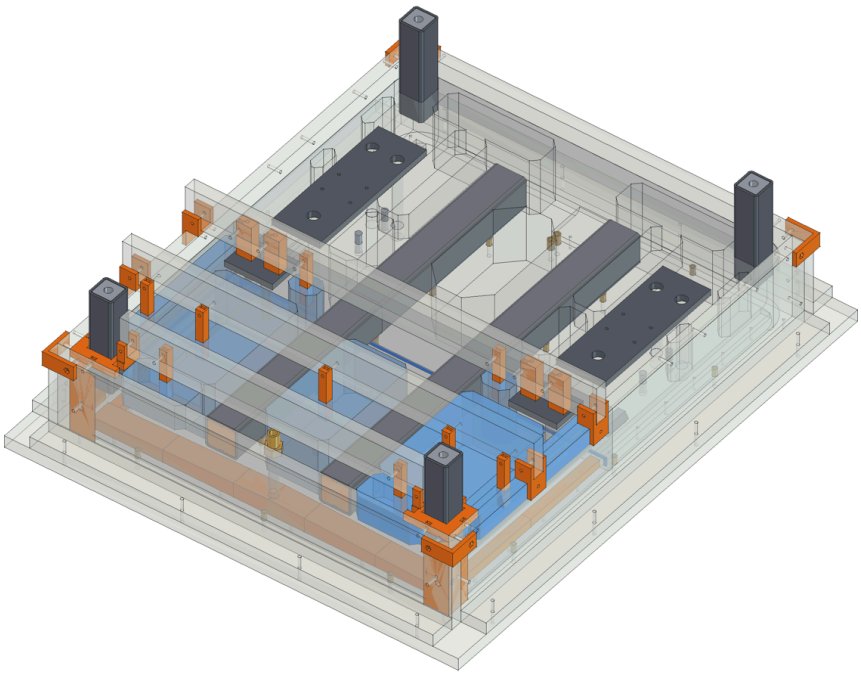


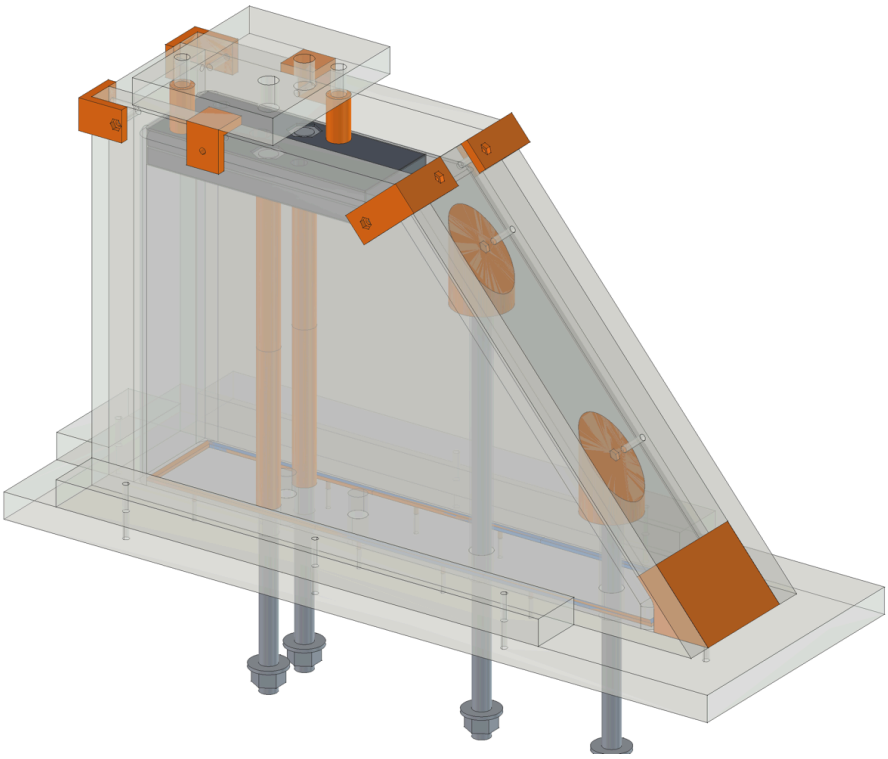


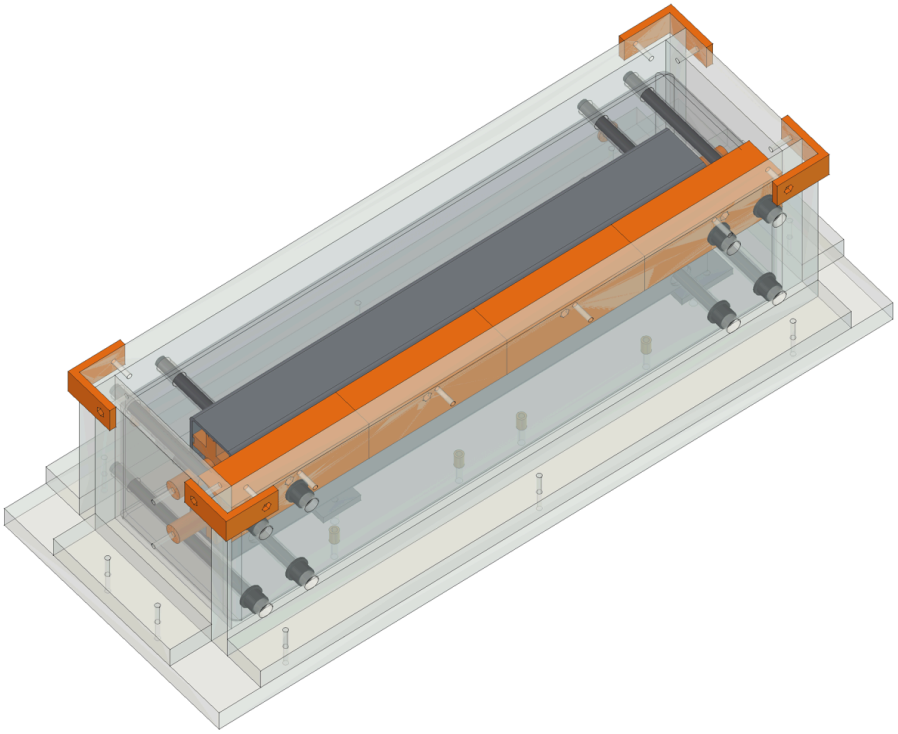
Locating the cast-in parts (the mold guide has the details; grease every screw and rod that comes out):

- **Inserts** (M5 enclosure, M5 riser hold-downs, M6 Y BK/BF): screwed to the floor from underneath, flush with the casting face.

- **Dowels** (rear half): two per upright, pushed halfway into holes in the floor; they stand 15 mm proud and are grouted into oversize holes in the upright's foot at assembly (section 4), so the two molds don't have to agree to a tenth.
- **Tie-rod holes and anchor bars:** greased 5/8 in rods up through the floor form the tie-rod holes right through the base. At the end of the pour the anchor bars go over the rods' tops, flush with the pour line, each screwed (M5) to printed hangers bolted to the bridges.
- **Draw-rod couplers** (rear half): on bolts through the partition.
- **Spigots** (rear half): through printed bushings in the partition, half their length out; the rear tubes, sleeves and end plugs slide onto them.
- **Legs:** through printed holders bolted into the corners above the pour line, each held at its cast-in depth by a pinch screw.
- **Cores:** taped XPS foam blocks on the wet 20 mm skin; a printed post glued into each bolts to the side of a bridge, which sets its height and stops it floating. They can stay in.
- **Uprights:** the top plate (both layers) hangs from the bridge on two greased 3/8 bolts. The two top tie-rod holes are printed sleeves, centered on greased 1/2 in rods pushed up from under the floor through the plate and the bridge; 5/8 in rods form the two slope holes (1.6 mm a side over the tie rods); a greased 1/2-13 bolt forms the two dowel holes (grouted at assembly).
- **Beam:** the tubes hang between printed locators bolted through the end walls (each leaves a 16 mm hole in the beam end). The locators only hold the tube ends; the 1/2 in rods through the tubes' 3/4 in holes set them along the beam, so a tube that's a little long or short doesn't matter. The rods also form the cap-screw holes (1.6 mm a side over the 3/8 in bolts). The support pads and riser inserts screw to the floor; butter each pad with thickened epoxy (epoxy and whiting, peanut-butter thick, about 3 mm) just before the tubes go in, so it fills the 1.5 mm gap to the front tube whatever the tube's size. The dowels sit in the bottom wall on printed stops.







3. Mixing and casting

Use the graded mix, by weight. The aggregate follows a Fuller-type packing curve (each size fills the gaps between the next size up), which is what lets the epoxy stay at 8.5 %:

Part	Size	Source	% of mix	For the whole frame
Epoxy		Raka 127 + 606	8.5 %	10.3 kg
Whiting (calcium carbonate)	325 mesh, under 0.05 mm	pottery supplier	9 %	10.5 kg
Fine sand	0.1-0.6 mm	pool filter sand (#20 silica)	15 %	10 kg (base) + 10 kg of 70 grit
Coarse sand	0.6-3 mm	concrete / all-purpose sand, sieved	27.5 %	19 kg (base) + 20 kg of #16 grit
Pea gravel	3-6 mm	bagged pea gravel, sieved	20 %	23 kg
Gravel	6-10 mm	3/8 in crushed stone (#8) or bigger pea gravel	20 %	23 kg

Two kitchen-type sieves (about 3 and 6 mm mesh, or hardware cloth) split the bagged sand and gravel into these bands. Wash anything dusty or clay-coated, and dry everything completely (spread it out in the sun, or an old oven tray): water stops the epoxy bonding. Crushed stone locks together a little stiffer than rounded gravel; rounded flows and packs a little better; either works.

Beam and uprights: aluminum oxide instead of the sand. Aluminum oxide grit is about five times stiffer than sand or granite. Brown fused #16 (about 1.2 mm) replaces the coarse sand and black 70 grit (about 0.2 mm, Harbor Freight) replaces the fine sand; together they make these castings roughly 35 % stiffer, about 17 % on the whole frame. The grit is heavier, so swap it by volume: **1.5 kg of #16 for every 1 kg of coarse sand, and 1.4 kg of 70 grit for every 1 kg of fine sand**, everything else (epoxy included) the same. A 50 lb bag of #16 and a 25 lb bag of 70 grit cover the beam, both uprights and a pair of test bars. The base keeps the sand. Don't over-vibrate these: the heavy grit can settle. Use an old paddle; the grit wears it.

8.5 % only works if the aggregate packs densely, so prove the mix before casting anything:

1. **Jar test.** Fill a 1 L jar with the dry aggregate blend (no whiting), shaking and tapping it down, then pour in water from a measuring cup until it reaches the top. The water is the void space. Nudge the coarse sand and fine sand up or down a few percent until it's as small as you can get it (below about 25 % of the jar is good).
2. **Test bars.** Print or build a 40 x 40 x 500 mm bar mold (waxed like the others) and cast two bars at 8.5 % and two at 9.5 %, about 2 kg each, vibrated the same way as the castings, plus two at 8.5 % with the alumina grits for the beam and uprights.
3. **Check them after 7 days.** Cut one bar of each pair open: there should be no voids larger than a grain of sand, and every stone should be wetted. Then tap-test the other: weigh it, rest it on two foam strips 112 mm in from each end, tap the middle with a small hammer and read the lowest strong peak on a phone spectrum app (phyphox, Spectroid). Its stiffness is about **$E\ (\mathrm{GPa}) = 4.8\mathrm{e}{-5} \times \mathrm{mass\ (kg)} \times \mathrm{frequency\ (Hz)}^2$**. The frame model assumes 30 GPa, about 570 Hz for a 1.9 kg bar; 25 GPa would cost about 13 % of the frame stiffness.
4. **Pick the mix.** Use 8.5 % if its bar is void-free and at least as stiff as the 9.5 % one; otherwise use 9.5 % and order the extra 1.5 qt epoxy kit. The alumina bar should come out about 35 % stiffer than the plain 8.5 % one (it's heavier, so use the formula, not the pitch); if it's well short, or it has voids, cast the beam and uprights with sand too.

Casting	Mix	Epoxy (8.5 %)
Base front	30.5 kg	2.6 kg
Base rear	41.9 kg	3.6 kg

Casting	Mix	Epoxy (8.5 %)
Uprights (alumina)	2 x 15.7 kg	2 x 1.2 kg
Beam (alumina)	26.3 kg	1.9 kg

Add 5-10 % for waste: about 11 kg, plus about 1 kg for the test bars, which fits the 3 gal kit (about 12.5 kg). Mix only what you can place; if you're running short, the 1.5 qt kit covers the gel coat.

1. **Dry and clean** all aggregate. Weigh everything out per batch before opening the epoxy. Keep batches to what you can place within about half the epoxy's pot life (10-15 kg each).
2. **Mix order:** epoxy resin + hardener (exact ratio, by weight), then the whiting until smooth, then the sands, then the gravel. Mix until every stone is wet and the color is even.
3. **Place in layers** of 30-40 mm. Vibrate each layer until the surface goes glossy and stops releasing bubbles, then add the next. Work the mix around the steel, inserts and cores with a stick. Vibrate the full mold for 20-30 minutes at the end.
4. **Level the open face** at the pour line, 25 mm below the wall tops: strike it off with a straight board in the two printed screed shoes (mold guide, "Pour"). Cover against dust, and leave it at 20-25 °C. Demold after the time your epoxy needs (usually 24-48 h), but don't load or machine anything for 7 days.
5. Pull all the greased screws and rods, the formers, fillers and plugs. Printed cores stay in; foam cores can be dug out or dissolved with a little acetone. Unbolt the mold kit; it's reused.
6. **Seal** the base top (the coolant floor) and the gutters with a thin epoxy gel coat. Paint the exposed leg ends.

Cast the rear base half first and the front half against it (above). Cast the two uprights together from one batch, before the beam, so you can check the top-plate spacing against the beam mold.